

Milne-Thomson Method

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Introduction

This method was developed by Louis Melville Milne-Thomson (1937). It helped greatly to simplify the process of finding an Analytic function whose real or imaginary part is given.



Louis Melville Milne-Thomson

Definition

- A function is said to be Analytic at a point if it is differentiable everywhere in some neighborhood of the point.
- A function is Analytic in a domain if it is analytic at every point in the domain.

Example

The function $f(z) = \frac{1}{(z-1)}$ is analytic at $z = 1 + i$.

Working Rule: To construct an Analytic Function by Milne Thomson Method

Case 1: When u is given

Let $f(z) = u(x, y) + iv(x, y)$ be analytic in a region D of $\mathbb{C}$.

Let $z = x + iy \in D$. So, $\bar{z} = x - iy$.

So, $x = \frac{z+\bar{z}}{2}$, $y = \frac{z-\bar{z}}{2i}$.

This relation may be regarded as a formal identity in z and $\bar{z}$.

Now, put $z = \bar{z}$.

So, $f(z) = u(z, 0) + iv(z, 0)$

Working Rule: To construct an Analytic Function by Milne Thomson Method (Continued)

Now, $f'(z) = u_x + iv_x = u_x - iu_y$. (By Cauchy-Riemann Equations)

Let $\phi_1(x, y) = u_x(x, y)$ & $\phi_2(x, y) = u_y(x, y)$.

Then $f'(z) = \phi_1(x, y) - i\phi_2(x, y) = \phi_1(z, 0) - i\phi_2(z, 0)$.

Integrating we get, $f(z) = \int (\phi_1(z, 0) - i\phi_2(z, 0))dz + k$.

Where k is an arbitrary constant

Working Rule: To construct an Analytic Function by Milne Thomson Method (Continued)

Case 2: When v is given

Similar as previous. Take $f'(z) = -if_y(x, y)$

Integrating, $f(z) = \int (\psi_1(z, 0) + i\psi_2(z, 0))dz + h.$

Where h is an arbitrary constant

Examples

Example 1 (When the real part is given)

If $u = x^2 - y^2$, find a corresponding analytic function.

Let $\phi_1(x, y) = u_x(x, y) = 2x$, $\phi_2(x, y) = u_y(x, y) = -2y$.

So, $\phi_1(z, 0) = 2z$, $\phi_2(z, 0) = 0$

Therefore, $f'(z) = \phi_1(z, 0) - i\phi_2(z, 0) = 2z$

So, $f(z) = z^2 + c$ is the required analytic function,

Where c is an arbitrary constant.

Examples (Continued)

Example 2 (When the imaginary part is given)

Show that $e^x(x\cos y - y\sin y)$ is a harmonic function. Find the analytic function for which $e^x(x\cos y - y\sin y)$ is the imaginary part.

Here, $v = e^x(x\cos y - y\sin y)$.

So, $v_x = e^x(x\cos y - y\sin y) + e^x \cos y$

$$v_{xx} = e^x(x\cos y - y\sin y + 2\cos y)$$

$$v_y = e^x(-x\sin y - y\cos y - \sin y)$$

$$v_{yy} = e^x(-x\cos y + y\sin y - 2\cos y)$$

$$\text{So, } v_{xx} + v_{yy} = 0$$

Hence, it is a harmonic function.

Examples (Continued)

let $\psi_1(x, y) = v_y(x, y)$, $\psi_2(x, y) = v_x(x, y)$

Therefore, $f'(z) = i(ze^z + e^z)$

So, $f(z) = i(ze^z) + d$, where d is an arbitrary constant

This is the required analytic function.

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Thank You!